# AI-HDL 2026: Design Phase 4 Documentation

**Phase Title:** Netlist to Chip Tapeout
**Duration:** April 9 – April 23, 2026
**Objective:** To take your Secure DP#3 optimized RTL design and turn it into a final, manufacturable physical layout (GDSII file).

## 1. Weekly Milestone Journey

This phase is structured as a 2-week sprint to ensure a stable foundation for the following optimization phases.

**Week 1 (Apr 9-15):** Synthesis, Floorplanning, Placement, CTS & Routing

- Run your final synthesis on the DP3 design.
- Create your chip's floorplan, including power grid and I/O pin placement.
- Run the main Placement, Clock Tree Synthesis (CTS), and Routing tools.
- This is a highly iterative process. You will run the tool, analyze timing, tweak settings, and run it again.

**Week 2 (Apr 16-22):** Sign-off Verification & Final Tape-out Package

- Run your final STA, DRC, and LVS checks.
- Your goal is "DRC/LVS Clean" with zero errors.
- This is the bar for a manufacturable chip.
- Package your final GDSII file.
- Write your final, comprehensive project report covering the entire 16-week journey (DP1-DP4).

## 2. Deliverables & Submission Guidelines

Teams must submit their work via a private GitHub repository, granting access to assigned mentors and judges.

**Required for DP4 Submission:**

- **Final GDSII File:** The manufacturable layout of your chip.
- **Sign-off Reports:** Clean DRC, LVS, and final STA reports.
- **Final Project Report:** A comprehensive report detailing your entire journey from DP1 to DP4, including your final PPA metrics, security features, and reflections on the process.
- **GitHub Tag**: A final commit tagged as "DP4-Submission".

## 3. Grading & Advancement Criteria

DP4 accounts for **30% of the total cumulative score**.

| Category | Weight | Criteria |
| --- | --- | --- |
| Physical Design Flow Execution | 30% | Successful, complete flow (Synth, P&R, CTS); no critical gaps. |
| Sign-off Quality | 20% | DRC/LVS clean reports. A manufacturable design. |
| Timing Closure & PPA | 20% | Final STA reports are clean; final PPA metrics are well-documented. |
| Submission Package | 20% | Complete GDSII, netlist, constraints, and final comprehensive report. |
| Presentation & Clarity | 10% | Clear repository organization and project story. |

## 4. Useful links

- AI-HDL website link: https://csm.arizona.edu/AIHDL
- AI-HDL Github link: https://github.com/prismlabarizona/AIHDL-2026
- Design phase 4 example: https://youtu.be/wDXgCwzYYIc
- Design phase 3 example: https://youtu.be/bSISaX4ymXg
- Design phase 2 example: https://www.youtube.com/watch?v=fUBtewf05rE
- Design phase 1 example: https://www.youtube.com/watch?v=M20AJiT_ETE&t=33s
- Webinar 3 – Base design Implementation: https://www.youtube.com/watch?v=rOaLiDxNm74
- RISC-V code repo: https://github.com/TinyTapeout/ttsky25a-tinyQV
- Peripheral template repo: https://github.com/prismlabarizona/AIHDL-2026/tree/master/tinyqv-full-peripheral-template
- Yosys Installation method: https://www.youtube.com/watch?v=XRexMO1B6s8&pp=2AYE
- OpenLane Installation method: https://www.youtube.com/watch?v=tFaTHGgpslA&pp=2AYB

**Important Date:** Milestone Review 4 (MR#4) is scheduled for **April 23, 2026.** Ensure your mentor check-ins are completed weekly to receive advisory feedback.